Exercise Set 2

Problem 1

a) Which of the expressions below correspond to the statement: "the probability of rain on Monday?"

1. $\Pr(\text{rain})$
2. $\Pr(\text{rain}|\text{Monday})$
3. $\Pr(\text{Monday}|\text{rain})$
4. $\Pr(\text{rain}, \text{Monday}) / \Pr(\text{Monday})$

b) Convert each statement into a probability expression. Which of the following statements corresponds to the expression: $\Pr(\text{Monday}|\text{rain})$?

1. The probability of rain on Monday.
2. The probability of rain, given that it is Monday.
3. The probability that it is Monday, given that it is raining.
4. The probability that it is Monday and that it is raining.

c) Which of the expressions below correspond to the statement: "the probability that it is Monday, given that it is raining?"

1. $\Pr(\text{Monday}|\text{rain})$
2. $\Pr(\text{rain}|\text{Monday})$
3. $\Pr(\text{rain}|\text{Monday}) \Pr(\text{Monday})$
4. $\Pr(\text{rain}|\text{Monday}) \Pr(\text{Monday}) / \Pr(\text{rain})$
5. $\Pr(\text{Monday}|\text{rain}) \Pr(\text{rain}) / \Pr(\text{Monday})$

Problem 2

Suppose there are two globes, one for Earth and one for Mars:

- Earth is 70% covered in water.
- Mars is 100% land.

Further suppose that one of these globes–you don't know which–was tossed in the air and produces a "land" observation. Assume that each globe was equally likely to be tossed. Show that the posterior probability that the globe was the Earth, conditional on seeing "land" ($\Pr(\text{Earth}|\text{land})$), is 0.23.

Problem 3

Consider the globe tossing example of McElreath. Compute and plot the grid approximate posterior distribution for each of the following sets of observations:

a) W, W, W. Use a uniform prior for p.

b) W, W, W, L. Use a uniform prior for p.

Use 8 water and 7 land tosses for questions 3c) to f):

c) Use a uniform prior for p.

d) Use a prior that is zero below $p = 0.5$ and a constant above $p = 0.5$.

e) Let us summarize the model output of 3d):

- How much posterior probability lies below p=0.55?
- How much posterior probability lies above p=0.75?
- 20% of the posterior probability lies below which value of p?
- Calculate the 90% *Highest posterior density interval (HPDI)* for p.

f) What difference does the better prior make? Compare the posterior distributions from 3c) and 3d) to the true value $p = 0.7$. Hint: Use *dens* of the rethinking package to create a density plot.

Problem 4

Modify the Metropolis algorithm code from the lecture slides to write your own simple MCMC estimator for globe tossing data.

References

McElreath, R. (2020). *Statistical rethinking- A Bayesian course with examples in R and Stan*. CRC press